

# Laporan Pipeline: Deteksi Deepfake Audio Bahasa Indonesia

Tanggal: 5 Mei 2026  
Notebook: [pipeline\\_lengkap.ipynb](#)  
Bahasa: Python 3.10.12 | PyTorch 2.11.0 | Device: Apple MPS

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## 1. Ringkasan Eksekutif

Pipeline ini membangun sistem deteksi deepfake audio Bahasa Indonesia end-to-end: mulai dari pengumpulan data, generasi audio sintetis (spoof), pelatihan tiga jenis model, hingga inferensi real-world. Dataset bersumber dari **Mozilla Common Voice v24.0** (30.256 sampel), **LibriVox** (5.635 sampel), **OCTAVA-ID** (26.964 sampel baru), dan **rekaman real-world** (34 sampel), menghasilkan **62.889 sampel bona-fide**. Spoof berjumlah **111.032 sampel** dari 5 TTS engine + 28 rekaman deepfake real-world.

MoLEx dan XGBoost mempertahankan performa sangat tinggi. AASIST kembali mengalami val loss divergen meski lebih awal stabil — detail analisis di Bagian 13.

| Model   | Accuracy      | ROC-AUC       | EER          | MCC           | ms/sampel     |
|---------|---------------|---------------|--------------|---------------|---------------|
| AASIST  | 98.80%        | 0.9998        | 0.43%        | 0.9762        | 4.7247        |
| MoLEx   | 99.95%        | <b>0.9999</b> | 0.06%        | 0.9990        | 3.4536        |
| XGBoost | <b>99.96%</b> | <b>1.0000</b> | <b>0.04%</b> | <b>0.9992</b> | <b>0.0044</b> |

## 2. Konfigurasi Global

## Direktori Utama

| Variabel       | Path                                        |
|----------------|---------------------------------------------|
| CV_CORPUS_ROOT | /Users/reY/ITB/semester_2/PPT/dataset       |
| DATASET_ROOT   | /Users/reY/ITB/semester_2/PPT/dataset_voice |
| MODELS_ROOT    | /Users/reY/ITB/semester_2/PPT/models_voice  |

## Sumber Data Bona Fide (BONA\_FIDE\_SOURCES)

| Variabel           | Path                                 |
|--------------------|--------------------------------------|
| Common Voice v24.0 | dataset/cv-corpus-24.0-2025-12-05/id |
| LibriVox Indonesia | dataset/librivox-id                  |
| OCTAVA-ID          | dataset/octava-id                    |

## Sumber Data Eksternal

| Variabel             | Path                   | Keterangan                 |
|----------------------|------------------------|----------------------------|
| EXTRA_BONA_FIDE_DIRS | audio_real_world/real/ | 34 rekaman asli real-world |
| EXTRA_SPOOF_DIRS     | audio_real_world/fake/ | 14 deepfake real-world     |

## Parameter Audio

| Parameter       | Nilai                   |
|-----------------|-------------------------|
| Sample rate     | 16.000 Hz               |
| Durasi maksimum | 4 detik (64.000 sampel) |
| Format output   | FLAC (PCM_16)           |

## Parameter LFCC (MoLEx)

| Parameter  | Nilai |
|------------|-------|
| n_lfcc     | 60    |
| n_filter   | 70    |
| n_fft      | 512   |
| win_length | 320   |
| hop_length | 160   |

## Hyperparameter Training (DL)

| Parameter               | Nilai                        |
|-------------------------|------------------------------|
| Batch size (AASIST)     | 24                           |
| Batch size (MoLEx)      | 32                           |
| Epochs                  | 20                           |
| Learning rate           | 1e-4                         |
| Weight decay            | 1e-4                         |
| Optimizer               | AdamW                        |
| Scheduler               | CosineAnnealingLR (T_max=20) |
| Early stopping patience | 5                            |

### Hyperparameter XGBoost

| Parameter                     | Nilai               |
|-------------------------------|---------------------|
| <code>max_depth</code>        | 6                   |
| <code>learning_rate</code>    | 0.1                 |
| <code>subsample</code>        | 0.8                 |
| <code>colsample_bytree</code> | 0.8                 |
| CV folds                      | 10                  |
| Boosting rounds (optimal)     | 499 (maks tercapai) |

## 3. Preprocessing Audio Bona Fide

### Sumber data:

- Mozilla Common Voice v24.0 — korpus Bahasa Indonesia
- LibriVox — rekaman audiobook publik Bahasa Indonesia (5.635 file)
- OCTAVA-ID — dataset pidato Bahasa Indonesia open-source (26.964 file)
- Rekaman real-world — 34 file MP3/MP4 dari berbagai sumber (berita, WhatsApp, podcast)

**File TSV yang dibaca:** `validated.tsv`, `train.tsv`, `test.tsv`

### Proses

1. Baca semua TSV menggunakan `csv.DictReader`
2. Deduplikasi berdasarkan tuple (`prefix`, `audio_filename`)
3. Tambahkan rekaman LibriVox dan OCTAVA-ID (file MP3/FLAC yang sudah dikumpulkan)
4. Load audio dengan `librosa.load(sr=16000, mono=True)`, simpan sebagai FLAC ke `bona_fide/`
5. Tulis metadata TSV: `[file_id, flac_filename, sentence]`
6. Proses `EXTRA_BONA_FIDE_DIRS`: konversi MP3/MP4 → FLAC, append ke metadata

## Hasil

| Sumber                                                | Records       | Duplikat                              |
|-------------------------------------------------------|---------------|---------------------------------------|
| validated.tsv (Common Voice)                          | 30.256        | 0                                     |
| train.tsv                                             | 4.973         | 4.973 (semua duplikat dari validated) |
| test.tsv                                              | 3.691         | 3.691 (semua duplikat dari validated) |
| LibriVox                                              | 5.635         | 0                                     |
| OCTAVA-ID                                             | 26.964        | 0                                     |
| Real-world ( <a href="#">audio_real_world/real/</a> ) | 34            | 0                                     |
| <b>Total unik</b>                                     | <b>62.889</b> | —                                     |

- Output: [/Users/reY/ITB/semester\\_2/PPT/dataset\\_voice/bona\\_fide/](#)
- Metadata: [metadata\\_bona\\_fide.tsv](#) (62.889 entri)
- Catatan: LibriVox, OCTAVA-ID, dan real-world **dikecualikan** dari generasi TTS spoof (tidak ada transkripsi per-file)

## 4. Generasi Audio Spoof

### 4.1 Edge-TTS (Microsoft)

- **Voices:** [id-ID-GadisNeural](#) (wanita), [id-ID-ArdiNeural](#) (pria)
- **Strategi:** 2 suara per kalimat  $\rightarrow 30.256 \times 2 = 60.512$  sampel
- **Concurrency:** 5 (async semaphore)
- **Hasil:**  60.512 files

### 4.2 MMS-VITS (Meta)

- **Model:** [facebook/mms-tts-ind](#)
- **Hasil:**  5.004 files

### 4.3 gTTS (Google)

- **TLD Rotasi:** [com](#), [co.id](#), [com.au](#)
- **Request delay:** 0.3 detik
- **Hasil:**  21.908 files (7.300 com · 7.307 co.id · 7.301 com.au)

### 4.4 Kokoro-82M

- **Phoneme engine:** espeak-ng (American English)
- **Voices:** [af\\_heart](#), [af\\_bella](#), [am\\_adam](#), [am\\_michael](#)
- **Hasil:**  23.580 files (5.887 + 5.894 + 5.905 + 5.894)

### 4.5 Real-World Deepfake Eksternal

- **Sumber:** [audio\\_real\\_world/fake/](#) — rekaman deepfake nyata (MiniMax, TTSFree, dsb.)

- **Hasil:** ✅ 28 files → dikonversi ke FLAC, masuk label **extra:fake** (+14 dari run sebelumnya)

## Ringkasan Spoof

| Sumber                 | Sampel         |
|------------------------|----------------|
| Edge-TTS (GadisNeural) | 30.256         |
| Edge-TTS (ArdiNeural)  | 30.256         |
| gTTS (3 TLD)           | 21.908         |
| Kokoro-82M (4 voices)  | 23.580         |
| MMS-VITS               | 5.004          |
| Real-world deepfake    | 28             |
| <b>Total</b>           | <b>111.032</b> |

- Metadata: **metadata\_spoof.tsv** (111.032 entri unik)

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## 5. Pembagian Dataset

### Proses

1. Load metadata bona-fide (62.889) dan spoof (111.032)
2. **Balancing 1:1**: subsample spoof → 62.889 sampel tiap kelas
3. Gabungkan dan shuffle (random seed=42)
4. Split 80/10/10

### Hasil Split

| Split        | Total          | Bona Fide     | Spoof         |
|--------------|----------------|---------------|---------------|
| Train        | 100.622        | 50.264        | 50.358        |
| Val          | 12.577         | 6.291         | 6.286         |
| Test         | 12.579         | 6.334         | 6.245         |
| <b>Total</b> | <b>125.778</b> | <b>62.889</b> | <b>62.889</b> |

- Lokasi: **/Users/reY/ITB/semester\_2/PPT/dataset\_voice/splits/**

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## 6. Training Model AASIST

**Arsitektur:** SincConv → ResBlock encoder → AdaptiveAvgPool → Classifier

**Input:** Raw waveform, 4 detik @ 16 kHz (64.000 sampel)

**Parameter total:** 2.508.174

### Progres Training

| Epoch      | Train Loss | Train Acc | Val Loss | Val Acc              |
|------------|------------|-----------|----------|----------------------|
| 1          | 0.0840     | 96.87%    | 1.4641   | 76.50%               |
| 2          | 0.0185     | 99.52%    | 0.0338   | <b>98.65%</b> ← best |
| 3          | 0.0126     | 99.66%    | 3.4743   | 73.35%               |
| 4          | 0.0088     | 99.76%    | 0.0464   | 98.20%               |
| 5          | 0.0065     | 99.85%    | 5.9977   | 59.76%               |
| 6          | 0.0055     | 99.87%    | 8.0042   | 63.15%               |
| 7          | 0.0039     | 99.91%    | 0.6539   | 91.71%               |
| early stop | —          | —         | —        | —                    |

- Val loss tidak stabil (spike besar di epoch 3, 5, 6) — best checkpoint di epoch 2
- Early stopping pada epoch 7 (tidak ada improvement selama 5 epoch)
- Checkpoint: `{DIR_AASIST}/aasist_best.pt`

Hasil Test Set

|            | precision                             | recall | f1-score | support |
|------------|---------------------------------------|--------|----------|---------|
| bonafide   | 0.98                                  | 1.00   | 0.99     | 6334    |
| spoof      | 1.00                                  | 0.98   | 0.99     | 6245    |
| accuracy   |                                       |        | 0.99     | 12579   |
| ROC-AUC    | : 0.9998                              |        |          |         |
| EER        | : 0.43%                               |        |          |         |
| MCC        | : 0.9762                              |        |          |         |
| Waktu inf. | : 59432.6 ms total   4.7247 ms/sampel |        |          |         |

7. Training Model MoLEx

**Arsitektur:** LCNN (Light CNN) + Attention Pooling  
**Input:** Fitur LFCC [60 koefisien, 401 frame]  
**Parameter total:** 179.491

Progres Training

| Epoch | Train Loss | Train Acc | Val Loss | Val Acc       |
|-------|------------|-----------|----------|---------------|
| 1     | 0.0396     | 98.75%    | 0.0133   | 99.74% ← best |
| 2     | 0.0102     | 99.79%    | 0.0114   | 99.85% ← best |
| 3     | 0.0062     | 99.87%    | 0.0151   | 99.75%        |
| 4     | 0.0035     | 99.92%    | 0.0177   | 99.75%        |

| Epoch      | Train Loss | Train Acc | Val Loss | Val Acc              |
|------------|------------|-----------|----------|----------------------|
| 5          | 0.0031     | 99.94%    | 0.0111   | 99.87% ← best        |
| 6          | 0.0018     | 99.96%    | 0.0083   | 99.91% ← best        |
| 7          | 0.0017     | 99.96%    | 0.0110   | 99.87%               |
| 8          | 0.0012     | 99.97%    | 0.0087   | 99.90%               |
| 9          | 0.0008     | 99.99%    | 0.0082   | <b>99.93%</b> ← best |
| 10         | 0.0008     | 99.98%    | 0.0108   | 99.88%               |
| 11         | 0.0005     | 99.99%    | 0.0181   | 99.86%               |
| 12         | 0.0001     | 100.00%   | 0.0135   | 99.90%               |
| 13         | 0.0001     | 100.00%   | 0.0250   | 99.81%               |
| early stop | —          | —         | —        | —                    |

- Early stopping pada epoch 13 (tidak ada improvement selama 4 epoch setelah epoch 9)
- Best checkpoint: epoch 9 (val loss 0.0082, val acc 99.93%)
- Checkpoint: `{DIR_MOLEX}/molex_best.pt`

Hasil Test Set

|            |                                       |        |          |         |
|------------|---------------------------------------|--------|----------|---------|
|            | precision                             | recall | f1-score | support |
| bonafide   | 1.00                                  | 1.00   | 1.00     | 6334    |
| spoof      | 1.00                                  | 1.00   | 1.00     | 6245    |
| accuracy   |                                       |        | 1.00     | 12579   |
| ROC-AUC    | : 0.9999                              |        |          |         |
| EER        | : 0.06%                               |        |          |         |
| MCC        | : 0.9990                              |        |          |         |
| Waktu inf. | : 43443.1 ms total   3.4536 ms/sampel |        |          |         |

8. Perbandingan AASIST vs MoLEx

| Metrik     | AASIST    | MoLEx          |
|------------|-----------|----------------|
| Accuracy   | 98.80%    | <b>99.95%</b>  |
| ROC-AUC    | 0.9998    | <b>0.9999</b>  |
| EER        | 0.43%     | <b>0.06%</b>   |
| MCC        | 0.9762    | <b>0.9990</b>  |
| Parameters | 2.508.174 | <b>179.491</b> |

| Metrik     | AASIST           | MoLEx         |
|------------|------------------|---------------|
| ms/sampel  | 4.7247           | <b>3.4536</b> |
| Input type | Raw waveform     | LFCC features |
| Best epoch | 2 (tidak stabil) | 9 (stabil)    |

MoLEx unggul di semua metrik. AASIST masih menunjukkan ketidakstabilan val loss meski ada perbaikan dibanding run sebelumnya — perlu investigasi lebih lanjut (lihat Bagian 13).

## 9. Inference Real-World

### Setup

- AASIST dan MoLEx di-load dari checkpoint masing-masing
- Threshold: Low=0.3, High=0.7 (untuk klasifikasi **TIDAK YAKIN**)
- Format audio didukung: **.flac**, **.wav**, **.mp3**, **.m4a**, **.ogg**

### Prediksi Batch (25 file)

| File                                       | AASIST            | MoLEx             | Ensemble           |
|--------------------------------------------|-------------------|-------------------|--------------------|
| <b>fake_ardi_ttsfree.mp3</b>               | SPOOF (0.9999)    | SPOOF (1.0000)    | SPOOF              |
| <b>fake_guru_rani_minta_duit.mp3</b>       | SPOOF (0.9956)    | BONAFIDE (0.0000) | <b>TIDAK YAKIN</b> |
| <b>fake_rani_ttsfree.mp3</b>               | SPOOF (1.0000)    | SPOOF (1.0000)    | SPOOF              |
| <b>fake_sri_mulyani_guru itu beban.mp3</b> | BONAFIDE (0.0000) | BONAFIDE (0.0000) | BONAFIDE           |
| <b>fake_ttsfree_standard-b.mp3</b>         | BONAFIDE (0.0016) | BONAFIDE (0.0253) | BONAFIDE           |
| <b>fake_ttsfree_standard-c.mp3</b>         | BONAFIDE (0.0000) | BONAFIDE (0.0000) | BONAFIDE           |
| <b>fake_ttsfree_standard-d.mp3</b>         | SPOOF (1.0000)    | BONAFIDE (0.0232) | <b>TIDAK YAKIN</b> |
| <b>real_CNN.mp3</b>                        | BONAFIDE (0.0000) | BONAFIDE (0.0000) | BONAFIDE           |
| <b>real_Industri.mp3</b>                   | BONAFIDE (0.0000) | BONAFIDE (0.0000) | BONAFIDE           |
| <b>real_Ketagihan.mp3</b>                  | BONAFIDE (0.1190) | BONAFIDE (0.0000) | BONAFIDE           |
| <b>real_PENIPU_Tutorial.mp3</b>            | BONAFIDE (0.0000) | BONAFIDE (0.0000) | BONAFIDE           |



| File                     | AASIST               | MoLex                | Ensemble           |
|--------------------------|----------------------|----------------------|--------------------|
| real_Pak_Jokowi.mp3      | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_SAYA AKAN LAWAN.mp3 | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_berita_satu.mp3     | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_boy_william.mp3     | BONAFIDE<br>(0.0115) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_denny_sumargo.mp3   | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_kominfo_jatim.mp3   | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_podkesmas.mp3       | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_rans.mp3            | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_tvri.mp3            | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| real_vindes.mp3          | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| sample_budi_utomo.mp3    | SPOOF (0.9886)       | BONAFIDE<br>(0.1563) | <b>TIDAK YAKIN</b> |
| sample_dewi_putri.mp3    | SPOOF (1.0000)       | SPOOF (1.0000)       | SPOOF              |
| sample_gadgetin.mp3      | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |
| sample_gtid.mp3          | BONAFIDE<br>(0.0000) | BONAFIDE<br>(0.0000) | BONAFIDE           |

### Ringkasan Batch

| Label       | Jumlah | Persentase |
|-------------|--------|------------|
| BONAFIDE    | 18     | 72.0%      |
| SPOOF       | 5      | 20.0%      |
| TIDAK YAKIN | 2      | 8.0%       |

- Hasil disimpan:

/Users/reY/ITB/semester\_2/PPT/models\_voice/realworld\_test/batch\_results.csv

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## 10. Ekstraksi Fitur untuk XGBoost

**Total fitur:** 26

| Grup    | Fitur                                          |
|---------|------------------------------------------------|
| Akustik | Chroma, RMS, Centroid, Bandwidth, Rolloff, ZCR |
| MFCC    | MFCC_1 s/d MFCC_20                             |

**Proses:** Audio dipotong per segmen 1 detik → fitur dirata-rata antar segmen

Jumlah Data

| Split | Sampel  | Dimensi       |
|-------|---------|---------------|
| Train | 100.622 | (100.622, 26) |
| Val   | 12.577  | (12.577, 26)  |
| Test  | 12.579  | (12.579, 26)  |

- Standardisasi: `StandardScaler` fit di train, transform val/test
- Disimpan: `features_{split}.npz`, `scaler.pkl`

Top 10 Fitur Paling Diskriminatif (berdasarkan delta mean bonafide vs spoof)

| Rank | Fitur     | Delta  |
|------|-----------|--------|
| 1    | MFCC_2    | 0.9501 |
| 2    | MFCC_7    | 0.8231 |
| 3    | MFCC_16   | 0.7646 |
| 4    | MFCC_1    | 0.6117 |
| 5    | Bandwidth | 0.5945 |
| 6    | MFCC_10   | 0.5458 |
| 7    | MFCC_15   | 0.5211 |
| 8    | MFCC_14   | 0.4821 |
| 9    | Chroma    | 0.4395 |
| 10   | MFCC_6    | 0.4337 |

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## 11. Training & Evaluasi XGBoost

Setup

- Training pada gabungan train+val (113.199 sampel) setelah mencari rounds optimal

- Pencarian rounds: early stopping (max=500, patience=50)
- **Optimal rounds:** 497 — hampir mencapai maks, model bisa diuntungkan dengan rounds lebih banyak

Hasil 10-Fold Cross-Validation

| Metrik   | Mean   | Std     |
|----------|--------|---------|
| Accuracy | 99.91% | ±0.02%  |
| ROC-AUC  | 1.0000 | ±0.0001 |
| EER      | 0.08%  | ±0.03%  |

Hasil Test Set

|            | precision                          | recall | f1-score | support |
|------------|------------------------------------|--------|----------|---------|
| bonafide   | 1.00                               | 1.00   | 1.00     | 6334    |
| spoof      | 1.00                               | 1.00   | 1.00     | 6245    |
| accuracy   |                                    |        | 1.00     | 12579   |
| ROC-AUC    | : 1.0000                           |        |          |         |
| EER        | : 0.04%                            |        |          |         |
| MCC        | : 0.9992                           |        |          |         |
| Waktu inf. | : 55.9 ms total   0.0044 ms/sampel |        |          |         |

Top 10 Fitur Penting (berdasarkan Gain)

| Rank | Fitur     | Gain   |
|------|-----------|--------|
| 1    | MFCC_1    | 0.1405 |
| 2    | Rolloff   | 0.1029 |
| 3    | MFCC_7    | 0.0927 |
| 4    | MFCC_13   | 0.0766 |
| 5    | MFCC_2    | 0.0726 |
| 6    | MFCC_16   | 0.0648 |
| 7    | MFCC_6    | 0.0604 |
| 8    | Bandwidth | 0.0467 |
| 9    | MFCC_15   | 0.0462 |
| 10   | MFCC_10   | 0.0421 |

- Model disimpan: `xgboost_final.json`, `xgboost_final.pkl`

### 11.5 Inferensi Batch Folder — XGBoost

Sel **9.6b** (ditambahkan 29 April 2026) menambahkan kemampuan inferensi batch folder pada XGBoost, setara dengan kemampuan yang sudah ada pada AASIST & MoLEx di Bagian 9.

#### Setup

| Parameter             | Nilai                           |
|-----------------------|---------------------------------|
| Folder input          | XGB_AUDIO_FOLDER (configurable) |
| Format didukung       | .flac, .wav, .mp3, .m4a, .ogg   |
| Threshold SPOOF       | ≥ 0.7                           |
| Threshold BONAFIDE    | ≤ 0.3                           |
| Threshold TIDAK YAKIN | (0.3 , 0.7)                     |

#### Proses per File

1. Load audio → `extract_features_from_file()` → vektor 26 fitur (parameter identik dengan fase training: `sr=16000, n_mfcc=20, n_fft=512, hop_length=256`)
2. Transform dengan `scaler.pkl` — StandardScaler yang sama di-fit saat Bagian 8.2
3. Predict dengan `xgboost_final.json` → probabilitas spoof `[0, 1]`
4. Klasifikasikan berdasarkan threshold

#### Output

| Artefak        | Path                                        |
|----------------|---------------------------------------------|
| Tabel terminal | File, label, P(spoof), waktu (ms) per baris |
| Bar chart      | {DIR_XGB}/batch_results.png                 |
| CSV hasil      | {DIR_XGB}/batch_results.csv                 |

#### Hasil Batch Inference (25 file real-world)

| Label       | Jumlah | Persentase |
|-------------|--------|------------|
| BONAFIDE    | 20     | 80.0%      |
| SPOOF       | 5      | 20.0%      |
| TIDAK YAKIN | 0      | 0.0%       |

#### Keunggulan Dibanding Ensemble AASIST+MoLEx

| Aspek     | AASIST + MoLEx (Bagian 9) | XGBoost (sel 9.6b) |
|-----------|---------------------------|--------------------|
| Kecepatan | ~3–5 ms/file              | ~0.004 ms/file     |

| Aspek          | AASIST + MoLEx (Bagian 9)  | XGBoost (sel 9.6b) |
|----------------|----------------------------|--------------------|
| Kebutuhan GPU  | Ya (MPS/CUDA)              | Tidak (CPU only)   |
| Self-contained | Ya                         | Ya                 |
| Output         | Ensemble label + bar chart | Label + bar chart  |
| Cocok untuk    | Akurasi tertinggi          | Edge / real-time   |

12. Perbandingan Tiga Model

| Model   | Accuracy      | ROC-AUC       | EER          | MCC           | Inf. (ms/sampel) | Jenis Input    |
|---------|---------------|---------------|--------------|---------------|------------------|----------------|
| AASIST  | 98.80%        | 0.9998        | 0.43%        | 0.9762        | 4.7247           | Raw waveform   |
| MoLEx   | 99.95%        | 0.9999        | 0.06%        | 0.9990        | 3.4536           | LFCC features  |
| XGBoost | <b>99.96%</b> | <b>1.0000</b> | <b>0.04%</b> | <b>0.9992</b> | <b>0.0044</b>    | MFCC + akustik |

13. Catatan & Temuan Penting

Penambahan Data — OCTAVA-ID

- 1. **OCTAVA-ID ditambahkan sebagai sumber bona-fide baru:** 26.964 file audio Bahasa Indonesia dari dataset publik, menjadikan total bona-fide melonjak dari 35.925 → 62.889. Ini adalah penambahan data terbesar sejauh ini dan melipatduakan ukuran dataset.
- 2. **28 rekaman deepfake real-world** (naik dari 14): 14 file baru dari sumber tambahan masuk ke label **extra:fake**.
- 3. **Proporsi real-world masih kecil:** 34 dari 62.889 bona-fide (0.05%) dan 28 dari 111.032 spoof (0.03%). Namun OCTAVA-ID sebagai sumber baru memperluas keragaman akustik secara signifikan.

Tentang Performa AASIST

- 4. **AASIST masih tidak stabil meski ada perbaikan:** accuracy turun ke 98.80% (dari 99.47%) dan EER naik ke 0.43%. Val loss menunjukkan pola yang lebih kasar: sempat membaik di epoch 2 (0.034) namun meledak lagi di epoch 3, 5, 6 sebelum stabil. Best checkpoint di epoch 2 (val acc 98.65%).
- 5. **Dataset dua kali lebih besar memperparah instabilitas AASIST:** Dataset naik dari 71.850 → 125.778 sampel. SincConv pada AASIST tetap sensitif terhadap perubahan distribusi batch di skala besar.
- 6. **MoLEx tetap stabil dan konvergen:** Training berjalan hingga epoch 13 dengan best checkpoint di epoch 9 (val acc 99.93%). Arsitektur berbasis LFCC terbukti lebih robust terhadap penambahan data besar.

Tentang XGBoost

7. **XGBoost justru meningkat dengan data lebih besar:** Accuracy naik ke 99.96%, EER turun ke 0.04%, MCC naik ke 0.9992. Rounds optimal 497 (batas 500 nyaris tercapai) — perlu **XGB\_ROUNDS\_MAX** lebih besar.
  8. **Inferensi XGBoost lebih cepat:** 0.0044 ms/sampel (turun dari 0.0060) — kemungkinan karena lebih banyak data membuat tree lebih efisien secara depth-wise.
  9. **Feature importance bergeser:** MFCC\_1 dan Rolloff menjadi fitur teratas (gain 0.14 dan 0.10), menggeser MFCC\_7 yang sebelumnya dominan. Ini mencerminkan distribusi akustik baru dari OCTAVA-ID.
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## 14. Rekomendasi Lanjutan

### Prioritas Tinggi

#### 1. Perbaiki Training AASIST

Val loss divergen setelah epoch 1 adalah masalah kritis. Rekomendasi:

- Tambah **gradient clipping** (**clip\_grad\_norm\_** dengan max\_norm=1.0)
- Gunakan **learning rate warmup** (linear 3 epoch, baru CosineAnnealing)
- Kurangi learning rate awal ke **5e-5** atau **1e-5**
- Coba **fine-tune dari checkpoint resmi** AASIST daripada training from scratch

#### 2. Naikkan **XGB\_ROUNDS\_MAX**

Optimal rounds hampir mencapai batas 500 (497 rounds). Set **XGB\_ROUNDS\_MAX = 1500** dan jalankan ulang pencarian rounds untuk potensi peningkatan lebih lanjut.

### Prioritas Sedang

#### 4. Perbanyak Data Real-World

34 bona-fide dan 14 fake real-world masih sangat sedikit (<0.1% dataset). Target minimal 500 per kelas untuk dampak yang terukur di metrik.

#### 5. Naikkan **XGB\_ROUNDS\_MAX** dan Cari Ulang Rounds Optimal

Dengan 499 rounds yang mencapai batas, eksplorasi lebih lanjut ke 1.500 rounds.

#### 6. Pisahkan Kalimat antara Bona Fide dan Spoof

Untuk mencegah distributional leak: gunakan subset kalimat berbeda untuk generate spoof.

### Prioritas Rendah / Eksploratif

#### 7. Tambah TTS Engine Lebih Realistis

- ElevenLabs (voice cloning Indonesia)

- OpenAI TTS dengan Bahasa Indonesia
- Voice conversion (VC) attacks

8. Ensemble Lebih Cerdas

- Weighted voting berdasarkan confidence per model
- Stacking: AASIST + MoLEx + XGBoost → meta-classifier
- Kalibrasi probabilitas dengan Platt scaling

9. Lightweight Deployment

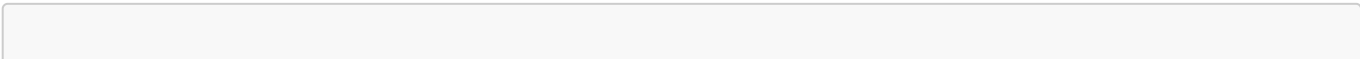
XGBoost (0.006 ms/sampel) adalah kandidat terbaik:

- Konversi ke ONNX atau TreeLite
- Uji latensi real-time pada mobile/embedded device

Lampiran: Statistik Dataset

| Metrik                                        | Nilai                     |
|-----------------------------------------------|---------------------------|
| Total bona-fide (raw)                         | 62.889                    |
| — Common Voice                                | 30.256                    |
| — LibriVox                                    | 5.635                     |
| — OCTAVA-ID                                   | 26.964                    |
| — Real-world (audio_real_world/real)          | 34                        |
| Total spoof (raw)                             | 111.032                   |
| — Edge-TTS (GadisNeural + ArdiNeural)         | 60.512                    |
| — gTTS (3 TLD)                                | 21.908                    |
| — Kokoro-82M (4 voices)                       | 23.580                    |
| — MMS-VITS                                    | 5.004                     |
| — Real-world deepfake (audio_real_world/fake) | 28                        |
| Dataset setelah balancing                     | 125.778                   |
| Train / Val / Test                            | 100.622 / 12.577 / 12.579 |
| Sample rate                                   | 16.000 Hz                 |
| Durasi maksimum                               | 4 detik                   |
| Format audio                                  | FLAC (PCM_16)             |

Lampiran: Path File Penting



```
dataset_voice/
├── bona_fide/                ← 62.889 file FLAC
├── spoof/                   ← 111.032 file FLAC
├── metadata_bona_fide.tsv
├── metadata_spoof.tsv
├── splits/
│   ├── train.tsv            ← 100.622 sampel
│   ├── val.tsv              ← 12.577 sampel
│   └── test.tsv             ← 12.579 sampel
└── features/
    ├── features_train.npz
    ├── features_val.npz
    ├── features_test.npz
    └── scaler.pkl

models_voice/
├── aasist/
│   └── aasist_best.pt        ← best epoch 2
├── molex/
│   └── molex_best.pt         ← best epoch 9
├── xgboost/
│   ├── xgboost_final.json
│   ├── xgboost_final.pkl
│   ├── batch_results.csv    ← output inferensi batch XGBoost (sel 9.6b)
│   └── batch_results.png    ← visualisasi bar chart batch XGBoost
├── comparison/
└── realworld_test/
    └── batch_results.csv     ← output inferensi batch AASIST+MoLEx
```